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(54) **A MALEATE SALT OF THE FREE BASE OF
N-[5-(AMINOSULFONYL)-4-METHYL-1,3-THIAZOL-2-YL]-N-METHYL-2-[4-(2-PYRIDINYL)-PHEN
YL]-ACETAMIDE, PHARMACEUTICAL FORMULATIONS, METHODS OF MANUFACTURE AND
USES THEREOF FOR THE TREATMENT OF HERPES VIRUSES**

MALEATSALZ DER FREIEN BASE VON
N-[5-(AMINOSULFONYL)-4-METHYL-1,3-THIAZOL-2-YL]-N-METHYL-2-[4-(2-PYRIDINYL)-PHEN
YL]-ACETAMID, PHARMAZEUTISCHE FORMULIERUNGEN, VERFAHREN ZUR HERSTELLUNG
UND VERWENDUNGEN DAVON ZUR BEHANDLUNG VON HERPESVIREN

SEL DE MALÉATE DE LA BASE LIBRE DE N- [5- (AMINOSULFONYL)
-4-METHYL-1,3-THIAZOL-2-YL] -N-METHYL-2- [4- (2-PYRIDINYL) -PHÉNYL] -ACÉTAMIDE,
FORMULATIONS PHARMACEUTIQUES, LEURS PROCÉDÉS DE FABRICATION ET LEURS
UTILISATIONS POUR LE TRAITEMENT DES VIRUS DE L'HERPES.

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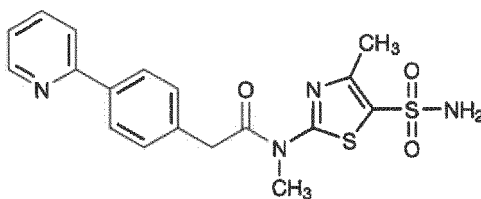
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BAY 57-1293
AIC316
Pritelivir

[0024] The term "maleate salt" or similar terms denote a maleate salt of the free base of the compound N-[5-(aminosulfonyl)-4-methyl-1,3-thiazol-2-yl]-N-methyl-2-[4-(2-pyridinyl)-phenyl]-acetamide.

[0025] In the context of the description the expression "maleate salt" or similar expressions denote the salt obtained from the free base of N-[5-(aminosulfonyl)-4-methyl-1,3-thiazol-2-yl]-N-methyl-2-[4-(2-pyridinyl)-phenyl]-acetamide when reacted with maleic acid. The maleate ion is the ionized form of maleic acid. Maleic acid or cis-butenedioic acid is an organic compound that is a dicarboxylic acid, a molecule with two carboxyl groups. Its chemical formula is HO₂CCHCHCO₂H. Maleic acid is the cis-isomer of butenedioic acid, whereas fumaric acid is the trans-isomer. Maleic acid is a less stable molecule than fumaric acid. Maleic acid is more soluble in water than fumaric acid. The melting point of maleic acid (135°C) is also much lower than that of fumaric acid (287°C). Both properties of maleic acid can be explained on account of the intramolecular hydrogen bonding that takes place in maleic acid at the expense of intermolecular interactions, and that are not possible in fumaric acid for geometric reasons. Specifically, maleic acid and the ionized form of maleic acid, i.e. the maleate ion, is further characterized by three double-bondings.

[0026] As used in this specification and in the appended claims, the singular forms of "a" and "an" also include the respective plurals unless the context clearly dictates otherwise.

The term "photostability" or similar expressions denote(s) the physico-chemical stability of the maleate salt of the free base N-[5-(aminosulfonyl)-4-methyl-1,3-thiazol-2-yl]-N-methyl-2-[4-(2-pyridinyl)-phenyl]-acetamide under light exposure of 300 - 800 nm wavelengths, and with a light exposure quantity of at least 1.2 million Lux hours, and with a light exposure energy of at least 200 watt hours/m² which does not result in pharmaceutically unacceptable change. Photostability may be tested in accordance with the ICH Topic Q1B guidance on Photostability Testing of New Active Substances and Medicinal Products, which is hereby incorporated by reference.

[0027] In the context of the present invention, the terms "about" and "approximately" denote an interval of accuracy that a person skilled in the art will understand to still ensure the technical effect of the feature in question. The term typically indicates a deviation from the indicated numerical value of $\pm 20\%$, preferably $\pm 15\%$, more preferably $\pm 10\%$, and even more preferably $\pm 5\%$.

[0028] It is to be understood that the term "comprising" is not limiting. For the purposes of the present invention the term "consisting of" is considered to be a preferred embodiment of the term "comprising of". If hereinafter a group is defined to comprise at least a certain number of embodiments, this is meant to also encompass a group which preferably consists of these embodiments only.

[0029] Furthermore, the terms "first", "second", "third" or "(a)", "(b)", "(c)", "(d)" etc. and the like in the description and in the claims, are used for distinguishing between similar elements and not necessarily for describing a sequential or chronological order. It is to be understood that the terms so used are interchangeable under appropriate circumstances and that the embodiments of the invention described herein are capable of operation in other sequences than described or illustrated herein.

[0030] In case the terms "first", "second", "third" or "(a)", "(b)", "(c)", "(d)" etc. relate to steps of a method or use there is no time or time interval coherence between the steps, i.e. the steps may be carried out simultaneously or there may be time intervals of seconds, minutes, hours, days, weeks, months or even years between such steps, unless otherwise indicated in the application as set forth herein above or below.

[0031] According to the present invention, the term "antiviral effective amount" means the total amount of each active component that is sufficient to show a meaningful patient benefit, i.e., healing of acute conditions characterized by inhibition of the herpes virus infection. When applied to an individual active ingredient, administered alone, the term refers to that ingredient alone. When applied to a combination, the term refers to combined amounts of the active ingredients that result in the therapeutic effect, whether administered in combination, serially or simultaneously.

[0032] In one aspect, the terms "prophylaxis or prevention" as used herein and in the claims refers to the administration or use of the herein disclosed compounds or compositions in order to protect e.g. a non-infected organism or e.g. a non-infected cell of an organism from being infected. With the context of the invention, however, this also means that an organism may be already infected by a virus, but the spread of said virus in the organism (from cell to cell) or within the organisms' social environment is prevented by the salts of the invention, particularly by a maleate salt of the invention.

The organism may be human or other mammal, whereby human is preferred. Thus, in one aspect of the invention, the organism to whom the compound or pharmaceutical composition is administered is a human being that is infected by a herpes virus, e.g., HSV-1 and/or HSV-2, or a human being that is in danger of being infected by such viruses. For further definitions on "prophylaxis/prevention" aspects of the invention see below.

[0033] The terms "treat, treating, treatment" as used herein and in the claims means preventing or ameliorating diseases associated with herpes virus infection.

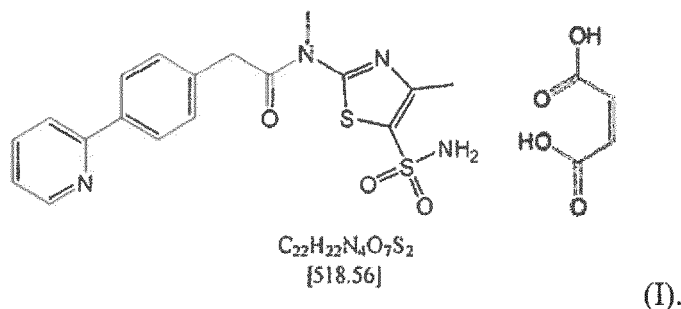
[0034] The physical characterization(s) of the salts, specifically of the maleate salt, as referred to herein, were performed using compendial methods as per European Pharmacopoeia (Ph. Eur.) and/or the U.S. Pharmacopeial Convention (USP).

[0035] Herein below, various embodiments of the invention are explained in more detail. Wherever respective alternatives in terms of ingredients in compositions, types of pharmaceutical compositions, concentrations of ingredients, periods of time of administration, frequencies of administration, medical indications to be treated are mentioned, the person skilled in the would immediately understand that individual combinations can be made as long as these are technically possible or if not otherwise explicitly indicated.

A maleate salt of the free base of pritelivir

[0036] In one embodiment the present application relates to a maleate salt of the free base of N-[5-(amino-sulfonyl)-4-methyl-1,3-thiazol-2-yl]-N-methyl-2-[4-(2-pyridinyl)phenyl]acetamide having the molecular formula $\text{HO}_2\text{CCHCHCO}_2\text{H}$. Characteristic properties of said maleate salt of the invention are shown in Figure 1. An Ortep Plot of the maleate salt is shown in Figure 2. Characteristic XRPD-peaks thereof are depicted in the Figures 10 and 11.

[0037] Said maleate salt is further characterized by a compound of the formula (I):



[0038] As used herein, the terms "storage stable" or "photostable" and corresponding terms indicate that the maleate salt of the invention does not decompose or decay for prolonged periods of time. This means that the maleate salt concentration remains constantly very high as can be measured using standard measuring methods permitting the identification of said compound and/or decomposition products thereof, e.g. HPLC, XRPD, and ^{13}C - and ^1H -NMR-spectroscopy methods showing the standard profile of the compound N-[5-(amino-sulfonyl)-4-methyl-1,3-thiazol-2-yl]-N-methyl-2-[4-(2-pyridinyl)phenyl]acetamide (i.e. pritelivir).

[0039] The physico-chemical characterisation of the salts, and particularly of the maleate salt as referred to herein was generally performed by using compendial methods as per European Pharmacopoeia (Ph. Eur.) and/or the U.S. Pharmacopeial Convention (USP).

[0040] Due to the above-mentioned general absence of impurities (either of the salts, specifically of the maleate salt as directly obtained from the manufacturing processes or due to decomposition of the N-[5-(amino-sulfonyl)-4-methyl-1,3-thiazol-2-yl]-N-methyl-2-[4-(2-pyridinyl)phenyl]acetamide), the purity of these salts, specifically of the maleate salt of the free base of pritelivir as the manufacturing product and upon formulation of compositions or medicaments comprising initially these salts of the free base of pritelivir is very high, i.e. a high concentration of N-[5-(amino-sulfonyl)-4-methyl-1,3-thiazol-2-yl]-N-methyl-2-[4-(2-pyridinyl)phenyl]acetamide per volume unit is achieved.

[0041] This is very beneficial when the salt of the invention is used in the formulation of pharmaceuticals and medicaments, specifically when used in topical formulations thereof, because the effective concentration of the active compound per volume unit is very high.

[0042] This allows reducing the amount of the maleate salt of the free base of pritelivir upon formulation whilst at the same time keeping a suitably high effective dose of the active compound. This is particularly desirable for pharmaceuticals used as topical compositions (unlike single unit dosage forms, e.g. essentially solid pharmaceuticals in form of tablets, etc., or - as another example - dosage forms for reconstitution in a pharmaceutically acceptable medium or carrier for immediate use as understood by the skilled person, e.g., for systemic or parenteral use), because the active ingredient in topically applied pharmaceuticals are subject to harsh environmental conditions in terms of changes in temperature, light exposure, such as UV-irradiation, (relative) humidity, mechanical stress upon application to affected areas of the

The patch may also contain other excipients such as cross-linkers, penetration enhancers, plasticizers, preservatives, antioxidants, fragrances, emollients. The backing layer may be transparent, semi-occlusive or occlusive, oxygen permeable, e.g., consisting of polyurethane ether or ester film, polyethylene, ethylene vinyl acetate or polyolephine film with a moisture vapor transmission rate (MVTR) from 50 to 3500 g/m²/day and a thickness from 20 to 150 μm. The backing layer should be very flexible and soft, transparent or colored and can be occlusive or perspiring, providing a masking effect of the cold sore. Moreover, it protects the damaged skin and the viral lesions from the external contact, thus reducing the patient's pain and the possibility of further contaminations or infections, and improving the re-epithelization process. The adhesive layer is protected from the external environment through a release liner, that has to be removed before applying the patch to the site of the body interested by the viral lesions. Once the patch is applied, through the self-adhesive layer, it can be kept on-site up to 6 to 24 hours, delivering the active ingredient into and across the skin.

[0089] In accordance with the invention, patches are prepared by a process comprising the steps of blending the solution of adhesive polymers in highly volatile solvents together with the other components and then casting the mixture on a silicone coated liner film, before drying and the final lamination. Highly volatile solvents evaporate, leaving the adhesive film on the release liner whereas the low volatile solvent remains in the adhesive layer preventing the drug crystallization. The polymers used according to this invention are those normally used to produce pressure sensitive adhesives (PSAs) or bio-adhesive film in an organic or aqueous solution, in a concentration ranging from 20 to 80 %, preferably from 20 to 50 % of the composition of the adhesive mixture, while the concentrations of the highly volatile solvent are from 10 to 50 %.

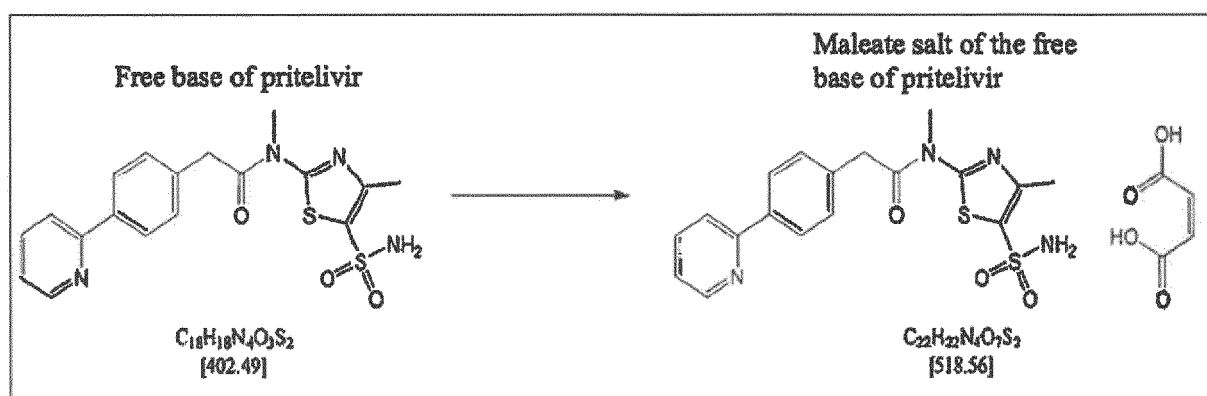
[0090] In accordance with the instant invention, other components of the adhesive layer or of the reservoir layer include thickening agents, chemical permeation enhancers, non-polymeric crystallization inhibitors, flavors, surfactants, cross-linkers, buffering agents, plasticizers, preservatives, anti-oxidants, pigments. The selected solvents and polymers must of course be compatible and form an homogeneous solution which may be uniformly casted. Low boiling point solvents, i.e. high volatile solvents with boiling point not higher than 100°C, are preferably water, ethanol, methanol, isopropyl alcohol, ethyl acetate, more preferably water.

[0091] Thus, per the invention, it is possible to produce an anti-herpes patch, having an effective amount of the maleate salt of the invention, as effective drug agent that can be continuously delivered to the site of application. The low volatile solvents helps to avoid crystallization by keeping dissolved the active substances in the matrix and affects the diffusivity of the drug through the matrix, to reach the skin and the site of action. The matrix must be chosen according to the physical-chemical properties of the low volatile solvents or the solvents blend. The polymer must provide a good cohesiveness to the final product. The quantitative composition of the adhesive blend is chosen in order to have an acceptable film in terms of thickness, cohesion properties, mechanical resistance, skin adhesion, peel properties and handling. The polymer blend range in the dry matrix is 5 to 50 %, most preferably 20 to 35 %, solubilized in a low boiling point solvent or solvent mixture. The solvent percentage ranges from 20 to 70 %, preferably from 35 to 55 %, in the mixture that has to be casted to produce the adhesive layer or the reservoir layer. In the dry matrix the amount of low boiling point solvent must not exceed 15 % by weight. The low volatile solvents, instead, are included in the dry matrix, entangled in the polymer and dissolving the active ingredients. The amount of these solvents in the dry state is in the range 10 to 50 %, particularly in the range of 30 to 55 %.

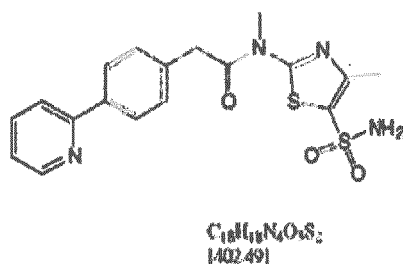
Processes for the manufacture of the salts of the invention

Maleate salt of the free base of pritelivir

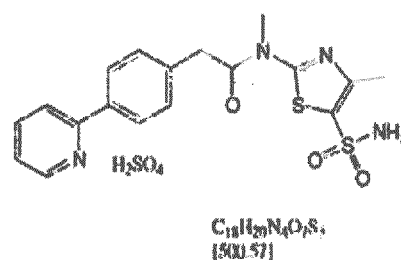
[0092]



Free base of pritelivir



Sulphate salt of the free base of pritelivir

**Synthesis scheme of a sulfate salt of the free base of pritelivir.****General process description:**

[0097] All equivalents, volumes (L/kg) and weights (kg/kg) refer to the weight of starting material free base of pritelivir. In a 2.5 L sulfonation flask with overhead stirring free base of pritelivir (1 eq., 1 wt., 250.0 g) is suspended in water (4.0 vol.) and ethanol (4.0 vol.). The suspension is heated to 50°C. A solution of sulphuric acid (96 %, 1.05 eq., 0.256 wt.) in water (0.256 vol.) is prepared and 15 % of the solution is added to the suspension at 50°C. After 5 minutes at 50°C the suspension converts to a thicker suspension and is stirred for further 150 minutes at 50°C. The remaining sulphuric acid solution is added at 50°C during 75 minutes to the suspension. The suspension cooled down to 21°C during 3 hours and stirred for 13 hours at said 21°C. The suspension is filtered and the filter cake is washed with a mixture of water (0.8 vol.) and ethanol (0.8 vol.). After deliquoring the filter cake on the filter the solid is transferred to a flask and dried on a rotovap (35°C, < 20 mbar, 6 h) to constant mass.

IPC: 93.59 % dry mass (160°C).

294.43 g sulfate salt of pritelivir free base (94.7 % yield not corrected for assay).

Characterisation of sulfate salt of free base of pritelivir:

[0098] Ethanol content by NMR: < 100 ppm (below quantification limit).

Water content by Karl Fischer titration: 6.33 % w/w (monohydrate 3.5 %, dihydrate 6.7 % w/w).

Assay by NMR: 95.3 % w/w as monosulfate (calculated monosulfate content by NMR higher than Karl Fischer titration allows).

76.6 % w/w as free base of pritelivir.

Sulfate salt: 18.1 % w/w, theoretical (monohydrate monosulfate): 18.9 % w/w (calculated, (difference free base) vs. (100 % - water content)).

19.4% w/w (calculated from elementary analysis and NMR assay (free base of pritelivir). Elementary analysis: atom found (% w/w) theoretical (% w/w)

C: 41.7	41.7
H: 4.5	4.3
N: 10.8	10.8
S: 17.8	18.6
O: 25.4	24.7

Specific process description for sulfate salt of pritelivir free base

[0099] Suspended free base of pritelivir in 1 L Ethanol and 1 L water is heated to a temperature of 50°C. 0.16 eq. sulfuric acid is added during 1 min. After approx. 5 min. a thick suspension is formed. Stirring for about further 2 hours follows to obtain a stirring suspension and 0.9 eq. sulfuric acid are added during 75 min. at temperature of 50°C. The suspension is allowed to cool down to room temperature and stirred for further 18 hours. Filtering, washing with 0.4 L ethanol/water 1/1 (v/v) and drying in vacuum follows.

Results:

[0100] Crude sulfate salt of pritelivir free base obtained: 294.4 g white solid (dry mass: 93.6 % (160°C), < 100 ppm Ethanol.

Sulfate content: pending (to be calculated from elementary analysis).

NMR: 85.7 % w/w calculated as monosulfate

TGA: 6.48% weight loss

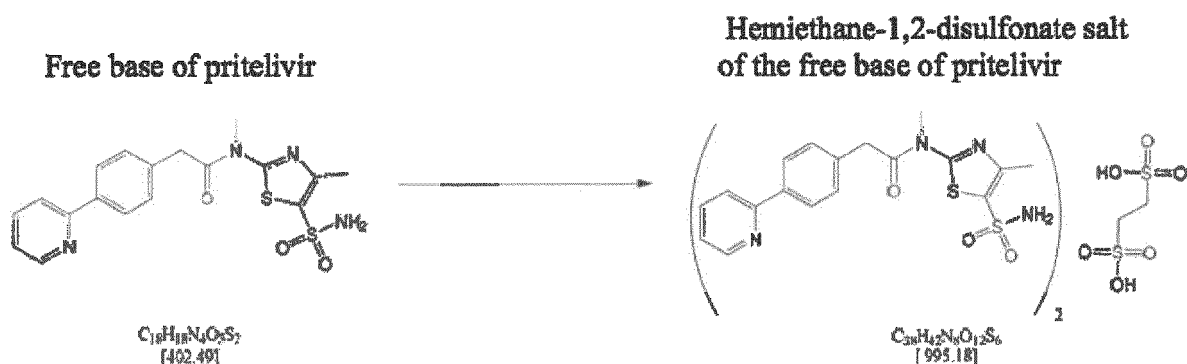
Karl Fischer: 6.23 % w/w water

Scale purity: 250 G

Product (% theory): 94.7 (uncorrected)

N-methyl-N-(4-methyl-5-sulfamoylthiazol-2-yl)-2-(4-pyridin-2-yl)phenylacetamide hemiethane-1,2-disulphonate salt of free base of pritelivir

Process for the manufacture of ethane-1,2-disulfonic acid salt of free base of pritelivir

[0101]

Synthesis scheme of the ethane-1,2-disulfonic acid salt of the free base of pritelivir.

General process description of the ethane-1,2-disulfonic acid salt of the free base of pritelivir

[0102] All equivalents, volumes (L/kg) and weights (kg/kg) refer to the weight of starting material free base of pritelivir. In a 2.5 L sulfonation flask with overhead stirring the free base pritelivir (1 eq., 1 wt., 250.0 g) is suspended in water (4.0 vol.) and ethanol (4.0 vol.). The suspension is heated to 53°C. A solution of ethane-1,2-disulfonic acid (0.60 eq., 0.312 wt.) in water (0.31 vol.) is prepared and 15 % of the solution is added to the suspension at 53°C. The suspension is seeded (0.02 % w/w) and after 45 minutes the addition of the acid solution continued during 10 minutes. The suspension is stirred for 90 minutes at 53°C and cooled down to 21°C during 3 hours and stirred for 13 hours at said 21°C. The suspension is filtered and the filter cake is washed with a mixture of water (0.8 vol.) and ethanol (0.8 vol.). After deliquoring the filter cake on the filter the solid is transferred to a flask and dried on a rotovap (35°C, < 20 mbar, 3 hours) to constant mass.

Results:

[0103] IPC: 96.84 % dry mass (160°C).

305.12 g crude ethane-1,2-disulfonic acid salt of the free base of pritelivir (98.7 % yield not corrected for assay).

Characterisation of the ethane-1,2-disulfonic acid salt of the free base of pritelivir

[0104] Ethanol content by NMR: not detectable.

Water content by Karl Fischer titration: 3.48 % w/w (monohydrate 1.8 %, dihydrate 3.5 % w/w).

Assay by NMR: 95.6 % w/w as hemisulfonate.

77.3 % w/w as free base.

Specific process description of the ethane-1,2-disulfonic acid salt of the free base of pritelivir

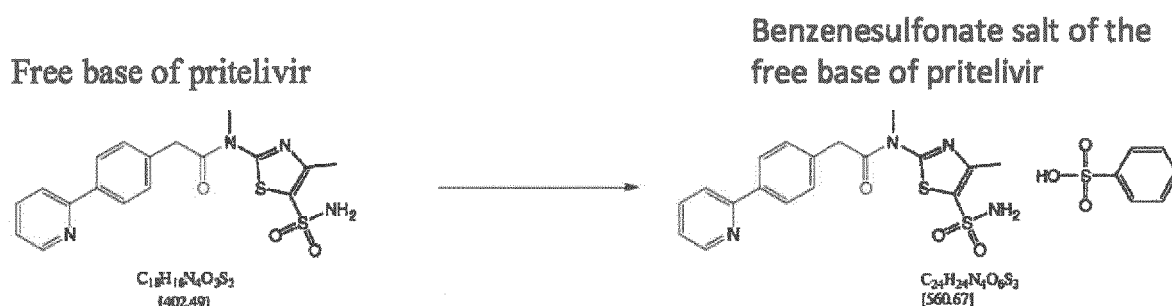
[0105] Free base of pritelivir is suspended in 1 L Ethanol and 1 L water and heated to temperature of 54°C. 0.09 eq. of ethane-1,2 disulfonic acid (11.6 g) are dissolved in 11.6 g water and filtered over syringe filter. Said ethane-1,2 disulfonic acid solution is added to free base suspension during 1 min. Seeds of ethane-1,2 disulfonic acid salt are added. Suspension is stirred for 40 min. at 54°C. 0.51 eq. ethane-1,2-disulfonic acid (66.4 g) is dissolved in water (66 g, filtered over syringe filter) during 1 hour at 54°C. 90 min. stirring follows at 54°C. The mixture is allowed to cool down to room temperature and stirred for further 18 hours at room temperature. Filtering, washing with ethanol/water 1/1 (v/v, 0.4 L) and drying in vacuum follows.

Results:

[0106] crude ethane-1,2 disulfonic acid salt: 305.1 g white solid (dry mass: 96.8 % (160°C), no ethanol in NMR visible. Ethane-1,2-disulfonic acid content: 0.53 eq. (NMR)
NMR: 95.6 % w/w calculated as hemi salt.
Karl Fischer: 3.48 % w/w water
Scale (purity): 250 g
Product (% theory): 98.7 (uncorrected)

N-methyl-N-(4-methyl-5-sulfamoylthiazol-2-yl)-2-(4-(pyridin-2-yl)phenyl)acetamide benzenesulfonate salt of free base of pritelivir**Process for the manufacture of benzene sulfonic acid salt of free base of pritelivir**

[0107]



Synthesis scheme of the benzene sulfonic acid salt of the free base of pritelivir.

General description of the process for manufacture of the benzene sulfonic acid salt of the free base of pritelivir

[0108] All equivalents, volumes (L/kg) and weights (kg/kg) refer to the weight of starting material pritelivir free base. In a 2.5 L sulfonation flask with overhead stirring, the free base of pritelivir (1 eq., 1 wt., 250.0 g) is suspended in water (4.0 vol.) and ethanol (4.0 vol.). The suspension is heated to 53°C. A solution of benzenesulfonic acid monohydrate (1.0 eq., 0.442 wt.) in water (0.15 vol.) is prepared and 15 % of the solution is added to the suspension at 50°C. The suspension is seeded (0.02 % w/w) and after 30 minutes the addition of the acid solution continues during 90 minutes. The suspension is cooled down to 21°C during 3 hours and stirred for 14 hours at 21°C. The suspension is filtered and the filter cake is washed with a mixture of water (0.8 vol.) and ethanol (0.8 vol.). After deliquoring the filter cake on the filter, the solid is transferred to a flask and dried on a rotovap (35°C, < 20 mbar, 4 h) to constant mass.

Results:

[0109] IPC: 99.63 % dry mass (160°C).
322.51 g crude benzene sulfonic acid salt of the free base of pritelivir (92.6 % yield not corrected for assay).

Characterisation of the benzene sulfonic acid salt of the free base of pritelivir

[0110] Ethanol content by NMR: 500 ppm

Water content by Karl Fischer titration: 0.17 % w/w (monohydrate 3.1 % w/w)

Assay by NMR: 102.1 % w/w as benzenesulfonate

73.4 % w/w as free base

Specific description of the process of benzene sulfonic acid salt of the free base of pritelivir

[0111] Free base of pritelivir is suspended in 1 L Ethanol and 1 L water and heated to a temperature of 52°C. 0.15 eq. benzene sulfonic acid (16.4 g) is added as solution in water (6.3 g) during 1 min. Seeds of benzene sulfonic acid salt of the free base of pritelivir are added and stirred for 30 min. at temperature of 52°C. 0.85 eq. benzene sulfonic acid (94.0 g) is added as a solution in water (30.5 g) during 90 min. This mixture is allowed to cool down to room temperature. Further stirring for about 18 hours at room temperature follows. Filtering, washing with 0.4 L ethanol/water 1/1 (v/v) and drying in vacuum follows.

Results:

[0112] crude benzene sulfonic acid salt of the free base of pritelivir : 322.5 g white solid (dry mass: 99.6 % (160°C) 500 ppm Ethanol

Benzene sulfonic acid content: 0.93 eq. (NMR)

NMR: 102.1 % w/w calculated as monosulfate

TGA: 0.15 % weight loss

Karl Fischer: 0.17 % w/w water

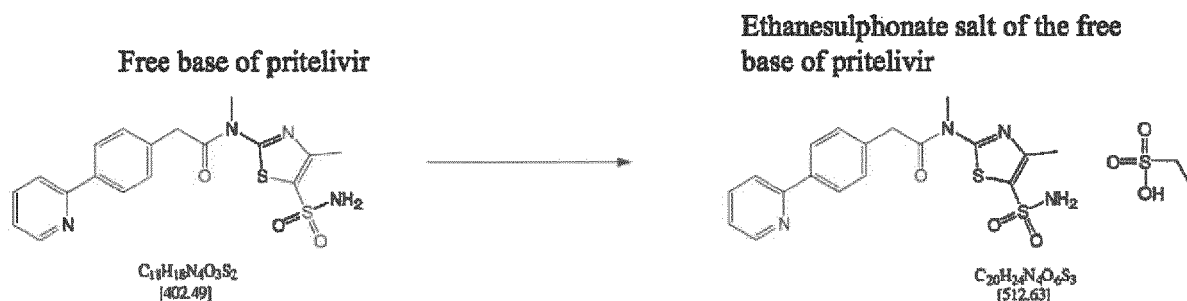
Scale (purity): 250 g

Product (% theory): 92.6 (uncorrected)

N-methyl-N-(4-methyl-5-sulfamoylthiazol-2-yl)-2-(4-(pyridin-2-yl)phenyl)acetamide ethanesulfonate salt of free base of pritelivir

Process for the manufacture of an ethane sulfonic acid salt of free base of pritelivir

[0113]



Scheme of the synthesis of an ethane sulfonic acid salt of free base of pritelivir.

General description of the process of manufacture of an ethane sulfonic acid salt of free base of pritelivir

[0114] All equivalents, volumes (L/kg) and weights (kg/kg) refer to the weight of starting material free base of pritelivir. In a 2.5 L sulfonation flask with overhead stirring free base of pritelivir (1 eq., 1 wt., 250.0 g) is suspended in water (4.0 vol.) and ethanol (4.0 vol.). The suspension is heated to 55°C. Ethanesulfonic acid (1.56 eq., 0.427 wt.) is dissolved in water (0.43 vol.) and added during 50 minutes. The suspension is allowed to cool down during 3 hours to 21°C (at around 38°C a thick suspension is may be formed) and further stirring for 14 hours at 21°C follows. The suspension is filtered and the filter cake is washed with a mixture of water (0.8 vol.) and ethanol (0.8 vol.). After deliquoring the filter cake on the filter, the solid is transferred to a flask and dried on a rotovap (35°C, < 20 mbar, 4 hours) to constant mass.

Results:

[0115] IPC: 96.31 % dry mass (160°C).

314.28 g crude ethane sulfonic acid salt of free base of pritelivir (98.7 % yield not corrected for assay).

FIG. 1:

Identification code	mo_P07102ACE1101_0m	
Empirical formula	$C_{22}H_{22}N_4O_7S_2$	
Formula weight	518.55	
Temperature	100(2) K	
Wavelength	0.71073 Å	
Crystal system	Monoclinic	
Space group	$C2/c$	
Unit cell dimensions	$a = 36.307(2)$ Å $b = 4.6551(3)$ Å $c = 29.3265(18)$ Å	$\alpha = 90^\circ$ $\beta = 115.0135(17)^\circ$ $\gamma = 90^\circ$
Volume	$4491.7(5)$ Å ³	
Z	8	
Density (calculated)	1.534 Mg / m ³	
Absorption coefficient	0.291 mm ⁻¹	
F(000)	2160	
Crystal size	$0.40 \times 0.02 \times 0.01$ mm ³	
Theta range for data collection	2.296 to 30.608°	
Index ranges	$-52 \leq h \leq 38$, $-6 \leq k \leq 6$, $-41 \leq l \leq 41$	
Reflections collected	26525	
Obs. Independent reflections	5414 [R(int) = 0.0388]	
Completeness to theta = 30.608°	99.0 %	
Absorption correction	Multi-scan	
Max. and min. transmission	0.997 and 0.914	
Refinement method	Full-matrix least-squares on F^2	
Data / restraints / parameters	6853 / 0 / 335	
Goodness-of-fit on F^2	1.028	
Final R indices [I > 2sigma(I)]	R1 = 0.0377, wR2 = 0.0882	
R indices (all data)	R1 = 0.0561, wR2 = 0.0961	
Largest diff. peak and hole	0.443 and -0.423 e·Å ⁻³	